STPM 2018 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $N_A = 6.02 \times 10^{23}$ mol^{-1}, $\varepsilon_0 = 8.85 \times 10^{-12}$ F m^{-1}, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}.

Section A [15 marks]

1. A metal sphere of radius 0.75 cm has a maximum electric field strength of 3.0×10^7 N C^{-1}. What is the charge on the surface of the metal sphere?

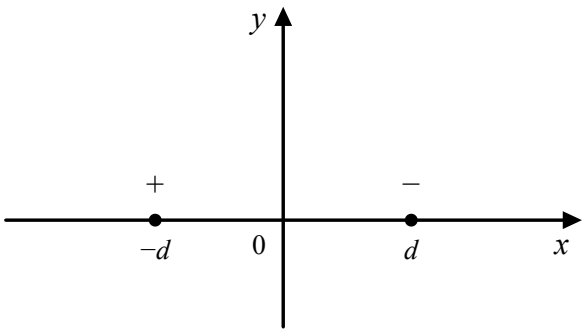
A 1.9×10^{-7} C B 2.5×10^{-5} C C 3.6×10^{19} C D 4.8×10^{21} C

Answer: A

$$Q = \frac{Er^2}{k} = \frac{(3.0 \times 10^7)(0.75 \times 10^{-2})^2}{9.0 \times 10^9} = 1.9 \times 10^{-7} \text{ C}$$

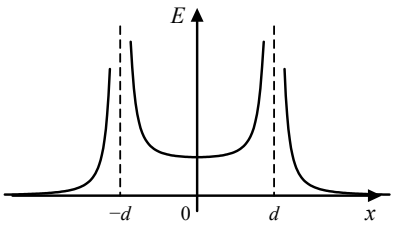
Explanation: The field is largest at the surface of the sphere.

2. Two oppositely charged particles of equal magnitude are placed at an equal distance d from the origin as shown in the diagram below.

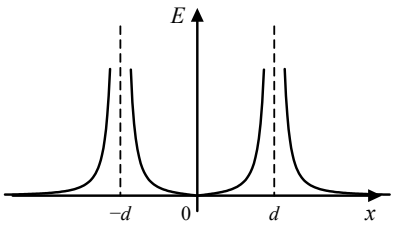


Which graph represents the variation of magnitude of electric field E along the x-axis?

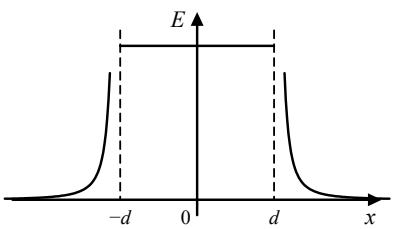
A



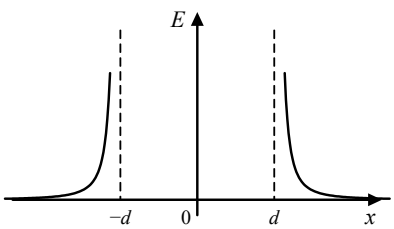
B



C



D



Answer: A

Near each charge E becomes very large. Between the charges the two fields point the same way and add, so E has a non-zero minimum at the origin; outside, E falls towards zero (A).

Explanation: For two like charges the field would be zero at the origin (B).

3. A capacitor with an air gap between the plates is connected to a battery. The battery is then removed from the capacitor. A dielectric is then used to fill in the air gap between the plates. Which statement is true about the capacitor with the dielectric?

A The energy stored in the capacitor increases.
B The charge stored in the capacitor decreases.
C The potential difference across the plates of the capacitor decreases.
D The electric field strength between the plates of the capacitor increases.

Answer: C

Isolated: Q constant, C increases, so $V = \frac{Q}{C}$ decreases (C)

Explanation: The field and the energy also decrease.

4. When a parallel plate capacitor is charged to a potential difference of 12.0 V, the charge on each plate is 2.16×10^{-4} C. If the potential difference across the capacitor is 30.0 V, what is the energy stored in the capacitor?

A 3.24×10^{-3} J B 6.48×10^{-3} J C 8.10×10^{-3} J D 1.62×10^{-2} J

Answer: C

$$C = \frac{2.16 \times 10^{-4}}{12.0} = 1.80 \times 10^{-5}\ \text{F};\ U = \frac{1}{2}(1.80 \times 10^{-5})(30.0)^2 = 8.10 \times 10^{-3}\ \text{J}$$

Explanation: C does not change with the p.d.

5. A potential difference of 12 V is supplied to a cable with a length of 50.0 cm. If the electric resistivity is $2.0 \times 10^{-8}\ \Omega$ m, what is the current density in the cable?

A $1.2 \times 10^{9}\ \text{A m}^{-2}$
B $2.1 \times 10^{6}\ \text{A m}^{-2}$
C $4.8 \times 10^{-7}\ \text{A m}^{-2}$
D $8.3 \times 10^{-10}\ \text{A m}^{-2}$

Answer: A

$$J = \frac{E}{\rho} = \frac{12/0.500}{2.0 \times 10^{-8}} = 1.2 \times 10^{9}\ \text{A m}^{-2}$$

Explanation: The cross-sectional area is not needed.

6. The drift velocity of electrons in an aluminium wire is 1.66×10^{-4} m s^{-1}. The cross-sectional area of the wire is 5.0×10^{-6} m^2. Assuming that the aluminium wire is supplied with one free electron by each aluminium atom, what is the current in the wire?
[Density of aluminium is 2700 kg m^{-3} and molar mass of aluminium is 0.027 kg mol^{-1}.]

A 0.010 A B 0.24 A C 4.0 A D 8.0 A

Answer: D

$$n = \frac{\rho N_A}{M} = \frac{(2700)(6.02 \times 10^{23})}{0.027} = 6.02 \times 10^{28}\ \text{m}^{-3}$$

$$I = nAev = (6.02 \times 10^{28})(5.0 \times 10^{-6})(1.60 \times 10^{-19})(1.66 \times 10^{-4}) = 8.0\ \text{A}$$

Explanation: One free electron per aluminium atom.

7. Which metal conductor and semiconductor correspond correctly to their characteristics at high temperature?

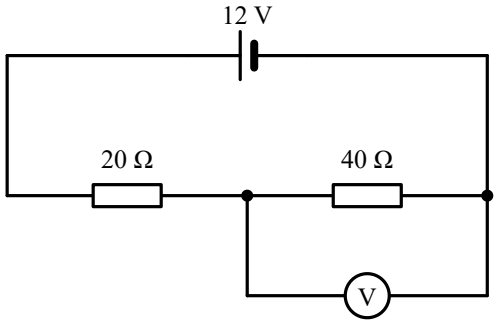
	Metal conductor	*Semiconductor*
A	The resistivity decreases	The resistivity increases
B	The number of charge carriers increases	The number of charge carriers increases
C	The conductivity of electricity increases	The conductivity of electricity decreases
D	The mean time between collisions decreases	The mean time between collisions decreases

Answer: D

At high temperature the lattice vibrates more in both, so the mean time between collisions decreases for both (D).

Explanation: The metal's resistivity increases; the semiconductor's carrier number increases so much that its resistivity decreases.

8. A circuit consists of a 12 V battery which is connected in series with two resistors, 20 Ω and 40 Ω, as shown in the diagram below.



A voltmeter of resistance 60 Ω is connected in parallel with the resistor of resistance 40 Ω. What is the reading of the voltmeter?

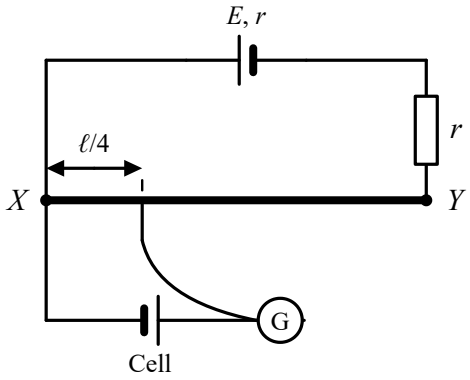
A 4.8 V B 6.5 V C 8.0 V D 9.0 V

Answer: B

$$40\ \Omega \parallel 60\ \Omega = 24\ \Omega;\ V = 12 \times \frac{24}{20 + 24} = 6.5\ \text{V}$$

Explanation: Without the voltmeter the p.d. would be 8.0 V.

9. A potentiometer consists of a battery of e.m.f. E with an internal resistance r, a resistor of resistance r, and a slide wire XY, as shown in the diagram below.



The length of the wire is ℓ and it has a resistance R. If the balanced length of the wire is $\frac{\ell}{4}$, what is the e.m.f. of the cell?

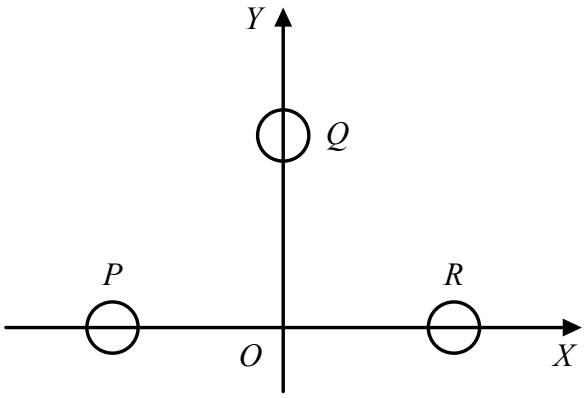
A $\frac{Er}{4(r+R)}$ B $\frac{ER}{4(r+R)}$ C $\frac{Er}{4(2r+R)}$ D $\frac{ER}{4(2r+R)}$

Answer: D

$$\text{Driver current } I = \frac{E}{r+r+R};\ \text{p.d. across } \frac{\ell}{4} = I\frac{R}{4} = \frac{ER}{4(2r+R)}$$

Explanation: The wire's resistance is shared uniformly along its length.

10. Three long parallel wires carry equal currents into or out of the page at points P, Q and R at equal distance from O, as shown in the diagram below.



Which currents in the wires will produce a magnetic field at O in the direction of OX if ⊗ shows the current flows into the page and ⊙ shows the current flows out of the page?

	P	Q	R
A	⊙	⊗	⊙
B	⊗	⊗	⊙
C	⊙	⊙	⊗
D	⊗	⊙	⊗

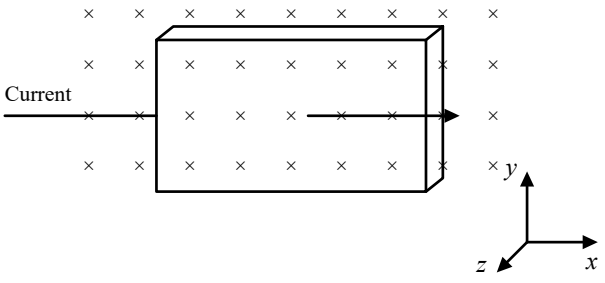
Answer: D

For a current out of the page the field circulates anticlockwise. At O: Q ⊙ gives $+X$; P ⊗ gives $-Y$ and R ⊗ gives $+Y$, which cancel.

Net field along OX: P ⊗, Q ⊙, R ⊗ (D).

Explanation: Use the right-hand grip rule for each wire.

11. A current which flows through a thin sheet of a conducting material placed in a uniform magnetic field is shown in the diagram below.



If the charge carriers in the material are electrons, what is the direction of Hall electric field in the conducting material?

A Positive y B Negative y C Positive z D Negative z

Answer: A

Electrons move opposite to the current ($-x$). With **B** into the page, $\mathbf{F} = -e\mathbf{v} \times \mathbf{B}$ pushes them to the top ($+y$) face, which becomes negative.

The Hall field points from the positive (lower) face to the negative (upper) face: positive y.

Explanation: For positive carriers the Hall field would be reversed.

12. The energy stored in an inductor is U_0 when a constant current flows through the inductor. While maintaining the cross-sectional area, the two ends of the inductor are pulled until the length of the inductor is twice the length of the original inductor. What is the energy stored in the inductor?

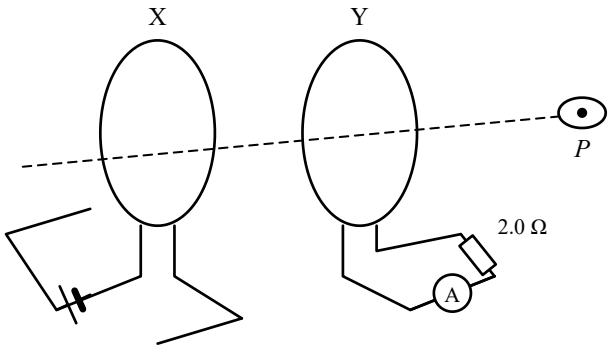
A $\frac{1}{8}U_0$ B $\frac{1}{4}U_0$ C $\frac{1}{2}U_0$ D U_0

Answer: C

$$L = \frac{\mu_0 N^2 A}{l}$$: doubling l halves L, so $U = \frac{1}{2}LI^2$ becomes $\frac{1}{2}U_0$

Explanation: The number of turns and the current are unchanged.

13. Two circular coils, X and Y, of negligible resistance are placed near to each other coaxially as shown in the diagram below.



The mutual inductance between the two coils is 2.0 H and the current in X is decreasing at a constant rate from 5.00 A to zero in 10 s. If Y is connected to a resistor of resistance 2.0 Ω, what are the magnitude and direction of induced current in Y as observed by P?

	Magnitude	*Direction*
A	0.50 A	Clockwise
B	0.50 A	Anti-clockwise
C	2.00 A	Clockwise
D	2.00 A	Anti-clockwise

Answer: B

$$\varepsilon = M\frac{dI}{dt} = (2.0)\frac{5.00}{10} = 1.0 \text{ V}; I = \frac{1.0}{2.0} = 0.50 \text{ A}$$

The flux from X through Y is decreasing, so the induced current in Y flows in the same sense as the current in X: anticlockwise as seen by P.

Explanation: Lenz's law: the induced current tries to maintain the decreasing flux.

14. A pure resistor is connected to an a.c. source with its current supply $I = 4.0 \sin 628t$ flowing in the circuit. Which statement is **not** true about the circuit?

A The r.m.s. current flowing through the resistor is 2.8 A.
B The maximum power dissipated from the resistor is 628 W.
C The peak value of the current flowing through the resistor is 4.0 A.
D The frequency of the current flowing through the resistor is 100 Hz.

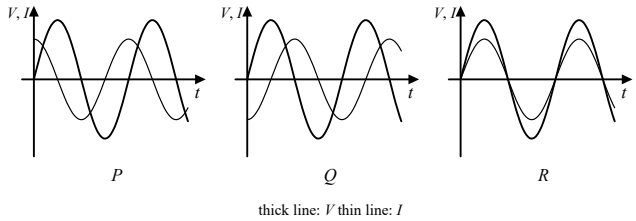
Answer: B

$$I_{rms} = \frac{4.0}{\sqrt{2}} = 2.8 \text{ A (A true)}; I_0 = 4.0 \text{ A (C true)}; f = \frac{628}{2\pi} = 100 \text{ Hz (D true)}$$

The power cannot be found without the resistance; 628 is the angular frequency, so B is not true.

Explanation: $P_{max} = I_0^2 R$.

15. A pure capacitor, a pure inductor and a pure resistor are connected separately to a sinusoidal voltage supply. The variation of the voltage V and current I with time t in each component is shown in the graphs P, Q and R below.



thick line: V thin line: I

Which graphs correspond correctly to their components?

	P	Q	R
A	Pure capacitor	Pure inductor	Pure resistor
B	Pure capacitor	Pure resistor	Pure inductor
C	Pure inductor	Pure capacitor	Pure resistor
D	Pure inductor	Pure resistor	Pure capacitor

Answer: A

In P the current leads the voltage by 90° (capacitor); in Q the current lags the voltage by 90° (inductor); in R they are in phase (resistor) (A).

Explanation: Compare where each curve reaches its maximum.

Section B [15 marks]

16. (a) Using Ampere's law, derive an expression for the magnetic flux density at a distance r from a long straight conductor carrying current I. [3]

Answer

Ampère's law: $\oint \mathbf{B}\cdot d\mathbf{l} = \mu_0 I$ — 1

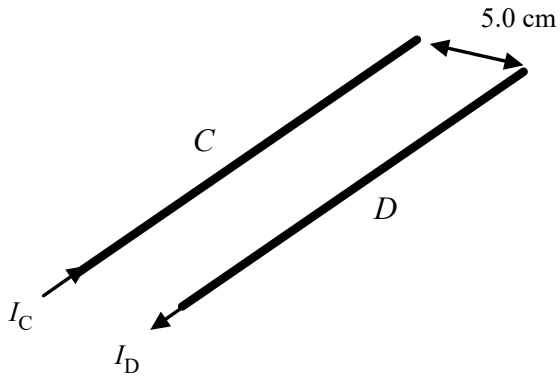
Take a circular path of radius r centred on the wire: B is tangential and constant, so $\oint \mathbf{B}\cdot d\mathbf{l} = B(2\pi r)$ — 1

$B(2\pi r) = \mu_0 I \Longrightarrow B = \dfrac{\mu_0 I}{2\pi r}$ — 1

3

Explanation: The field lines are concentric circles around the wire.

(b) Two long parallel wires, C and D, are separated by a distance of 5.0 cm as shown in the diagram below.



The currents in C and D are 2.0 A and 9.3 A respectively.

(i) Derive an expression for the force per unit length acting on D due to the current in C. [2]

Answer

Field of C at D: $B_C = \dfrac{\mu_0 I_C}{2\pi r}$ — 1

Force on length L of D: $F = B_C I_D L \Longrightarrow \dfrac{F}{L} = \dfrac{\mu_0 I_C I_D}{2\pi r}$ — 1

2

Explanation: D is perpendicular to the field of C.

(ii) Determine the magnitude and direction of the force per unit length on D. [2]

Answer

$\frac{F}{L} = \frac{(4\pi \times 10^{-7})(2.0)(9.3)}{2\pi(0.050)} = 7.44 \times 10^{-5}$ N m^{-1} 1

The currents are in opposite directions, so the force is repulsive: on D it acts away from C 1

2

Total for question **7**

Explanation: The same force acts on C in the opposite direction.

17. An a.c. source with r.m.s. voltage of 220 V is connected in series with a resistor of resistance 81 Ω and a capacitor of capacitance 22 μF. The r.m.s. current in the circuit is 1.8 A. Calculate

(a) the impedance of the circuit, [2]

Answer

$Z = \frac{V_{\text{rms}}}{I_{\text{rms}}} = \frac{220}{1.8}$ 1

$Z = 122\ \Omega$ 1

2

Explanation: Impedance = r.m.s. voltage ÷ r.m.s. current.

(b) the reactance of the capacitor, [2]

Answer

$X_C = \sqrt{(Z^2 - R^2)} = \sqrt{(122^2 - 81^2)}$ 1

$X_C = 91.5\ \Omega$ 1

2

Explanation: $Z^2 = R^2 + X_C{}^2$.

(c) the frequency of the source, [2]

Answer

$f = \frac{1}{2\pi C X_C} = \frac{1}{2\pi(22 \times 10^{-6})(91.5)}$ 1

$f = 79$ Hz 1

2

Explanation: $X_C = 1/(2\pi fC)$.

(d) the phase angle between voltage and current. [2]

Answer

$\tan \varphi = \frac{X_C}{R} = \frac{91.5}{81} \Longrightarrow \varphi = 48.5°$ 1

The voltage lags the current 1

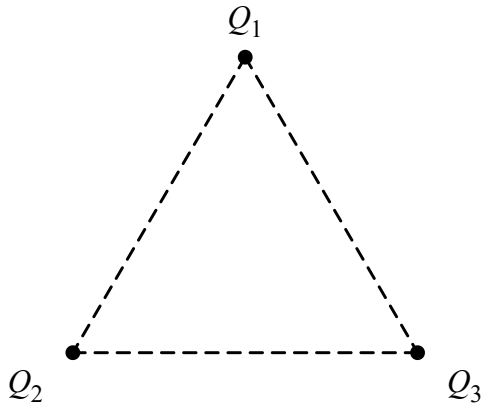
2

Total for question **8**

Explanation: In an *RC* circuit the current leads.

Section C [30 marks]

18. (a) Three charged particles in the air are located at the corners of an equilateral triangle as shown in the diagram below.



The charges Q_1, Q_2 and Q_3 are +1.0 μC, +100.0 μC and +100.0 μC respectively. The distance between the particles is 1.0 cm.

(i) Determine the magnitude and direction of the net force on Q_1. [4]

Answer

Force from each of Q_2, Q_3: $\frac{(9.0 \times 10^9)(1.0 \times 10^{-6})(100.0 \times 10^{-6})}{(0.010)^2} = 9.0 \times 10^3$ N 1

Both are repulsions along the sides, each at 30° to the vertical through Q_1; the horizontal components cancel 1

$F = 2(9.0 \times 10^3) \cos 30° = 1.56 \times 10^4$ N 1

Directed perpendicular to Q_2Q_3, away from it (vertically upwards) 1

4

Explanation: By symmetry the net force is along the line of symmetry.

(ii) Calculate the percentage of the electrostatic force reduced when the air is replaced by water with a relative permittivity of 80.0. [3]

Answer

In water $F' = \frac{F}{\varepsilon_r} = \frac{F}{80.0}$ 1

Reduction $= (1 - \frac{1}{80.0}) \times 100\%$ 1

$= 98.8\%$ 1

3

Explanation: The force in water is 195 N.

(iii) Based on your answer in (ii), explain why cooking salts dissolve in water easily. [3]

Answer

Salt is made of oppositely charged ions (Na^+ and Cl^-) held by electrostatic attraction 1

In water this attraction is reduced to 1/80 of its value in air 1

so the ions are easily separated by the water molecules and thermal motion: the salt dissolves 1

3

Explanation: Water has a very high relative permittivity.

(b)

(i) Define *Gauss's law*. [1]

Answer

The total electric flux through a closed surface equals the net charge enclosed divided by ε_0:

$\Phi = \frac{Q}{\varepsilon_0}$ 1

1

Explanation: The closed surface is a Gaussian surface.

(ii) A thin conducting square plate with a length of 10.0 cm in free space has total charge +60.0 μC. Determine the magnitude and direction of the electric field. [4]

Answer

The charge spreads over both faces of the plate: 30.0 μC on each face, area 0.0100 m^2 1

$\sigma = \frac{30.0 \times 10^{-6}}{0.0100} = 3.0 \times 10^{-3}$ C m^{-2} 1

$E = \frac{\sigma}{\varepsilon_0} = \frac{3.0 \times 10^{-3}}{8.85 \times 10^{-12}} = 3.4 \times 10^{8}$ N C^{-1} 1

Perpendicular to the plate, pointing away from it on both sides 1

4

Total for question **15**

Explanation: Close to the plate, away from its edges.

19. (a) State **both** Kirchhoff's laws. [2]

Answer

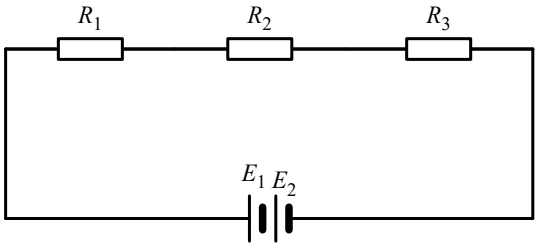
First law: the sum of currents entering a junction equals the sum of currents leaving it 1

Second law: around any closed loop, the sum of the e.m.f.s equals the sum of the *IR* products 1

2

Explanation: Conservation of charge and of energy.

(b) Two batteries of e.m.f. E_1 and E_2, each with negligible internal resistance, are connected in series with three resistors of resistance R_1, R_2 and R_3, as shown in the diagram below.



(i) Using Kirchhoff's law, deduce an expression for the current which flows in the circuit. [2]

Answer

Round the loop: $E_1 + E_2 = IR_1 + IR_2 + IR_3$ 1

$I = \frac{E_1 + E_2}{R_1 + R_2 + R_3}$ 1

2

Explanation: The two cells act in the same direction.

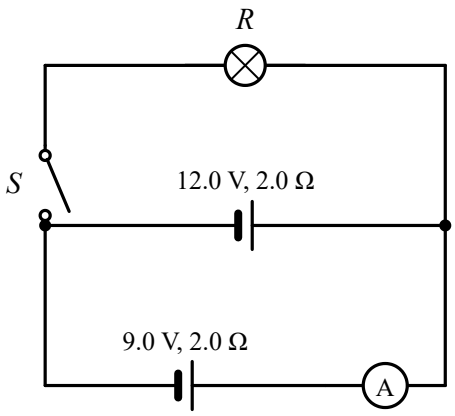
(ii) Show that Kirchhoff's law is consistent with the Principle of the Conservation of Energy. [4]

Answer

Multiply the loop equation by I: $IE_1 + IE_2 = I^2R_1 + I^2R_2 + I^2R_3$	1
IE is the power supplied by each cell (rate of conversion of chemical energy)	1
I^2R is the power dissipated as heat in each resistor	1
Power supplied = power dissipated, so energy is conserved	1
	4

Explanation: Kirchhoff's second law is the circuit form of energy conservation.

(c) Two dry cells of e.m.f.s 12.0 V and 9.0 V respectively, each with an internal resistance of 2.0 Ω, are connected in parallel. The cells are connected in series with a switch S and a bulb of resistance R, as shown in the diagram below.



(i) When S is opened, determine the current measured by the ammeter. [2]

Answer

With S open the two cells form a single loop and oppose each other	1
$I = \frac{12.0 - 9.0}{2.0 + 2.0} = 0.75$ A	1
	2

Explanation: The 9.0 V cell is being charged.

(ii) When S is closed, the reading of the ammeter is zero. Determine the resistance of the bulb. [3]

Answer

No current in the 9.0 V cell, so the p.d. across the parallel branches equals its e.m.f.: 9.0 V 1

Current from the 12.0 V cell = $\frac{12.0 - 9.0}{2.0}$ = 1.5 A, all of which flows through the bulb 1

$R = \frac{9.0}{1.5} = 6.0\ \Omega$ 1

3

Explanation: The 9.0 V cell is then exactly balanced.

(iii) Calculate the power dissipated by the bulb. [2]

Answer

$P = I^2R = (1.5)^2(6.0)$ 1

$P = 13.5$ W 1

2

Total for question **15**

Explanation: Equivalently $V^2/R = 9.0^2/6.0$.

20. (a) Explain self-induction. [3]

Answer

A changing current in a coil produces a changing magnetic flux linking the coil itself 1

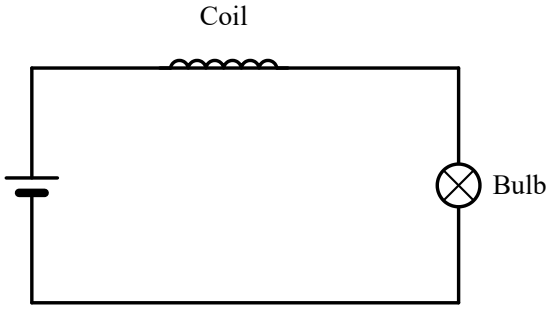
This induces an e.m.f. in the same coil (Faraday's law) 1

By Lenz's law the e.m.f. opposes the change in current: $\varepsilon = -L\frac{dI}{dt}$ 1

3

Explanation: It is called a back e.m.f.

(b) A light bulb which glows with a steady brightness in a circuit containing a coil is shown in the diagram below.



Explain the changes in the brightness of the bulb that could occur when an iron core is inserted into the coil. [3]

Answer

Inserting the iron core greatly increases the magnetic flux linkage (and the inductance) of the coil	1
The changing flux induces a back e.m.f. that opposes the current, so the bulb dims momentarily	1
Once the core is still, the current returns to its steady value (set by the resistance), so the bulb returns to its original brightness	1
	3

Explanation: With a steady d.c. supply the inductance affects only changes in current.

(c) A current changing at a rate of 10.0 A s^{-1} flows through a coil of 500 turns and produces a back e.m.f. of 24.5 mV. Determine the

(i) self-inductance of the coil, [2]

Answer

$L = \frac{\varepsilon}{\mathrm{d}I/\mathrm{d}t} = \frac{24.5 \times 10^{-3}}{10.0}$	1
$L = 2.45 \times 10^{-3}$ H	1
	2

Explanation: Back e.m.f. = L × rate of change of current.

(ii) magnetic flux through each turn when the current is 4.00 A. [2]

Answer

$N\Phi = LI \Longrightarrow \Phi = \frac{(2.45 \times 10^{-3})(4.00)}{500}$	1
$\Phi = 1.96 \times 10^{-5}$ Wb	1
	2

Explanation: LI is the total flux linkage.

(d) A coil with inductance of 2.0 H, resistance of 5.0 Ω and a switch are connected in series to a 12.0 V battery of negligible internal resistance. The switch is closed and the current in the circuit builds up until it becomes steady. At the instant when the current is 1.40 A, determine the

(i) back e.m.f. in the coil, [3]

Answer

Kirchhoff: $12.0 = IR$ + back e.m.f.	1
Back e.m.f. = $12.0 - (1.40)(5.0)$	1
= 5.0 V	1
	3

Explanation: The back e.m.f. falls as the current grows.

(ii) rate of change of current in the coil. [2]

Answer

$\frac{dI}{dt} = \frac{\varepsilon}{L} = \frac{5.0}{2.0}$	1
$= 2.5\ \text{A s}^{-1}$	1
	2
Total for question	**15**

Explanation: The final steady current is 12.0/5.0 = 2.4 A.